

HOUSING PLANS

The role of thermal dissipation falls solely on the trusty old cabinet, which until yesterday was merely considered a necessary evil...

“Do I really need a new cabinet?” Well, the fact is, if you asked us this a few months before the Pentium 4s and Athlons of the world came along, we’d have answered in the negative! Processors, memory, graphics solutions, and hard drives have all become significantly faster. The downside is power consumption, which results in higher heat dissipation. A few years ago, a 180-watt power supply (we’re talking real output and not claimed) was sufficient to power an entire PC. Today’s graphics cards happily consume in excess of 200 watts, and top-end dual-core processors demand close to 150 watts!

All this heat generated has to go someplace, and that’s why choosing the right cabinet has become essential for a silent, stable computing experience.

MYTHS AND REALITIES

Combos are better: a huge mistake most people make is to settle for the cabinet and power supply combo their hardware vendor dishes out. Cabinets are available discretely, and you can also select a power supply—even if it’s a different brand. There are no compatibility issues, and neither are you under any compul-

sion to buy them as a bundle. You may need a snazzy-looking cabinet but a cheaper power supply. A hardcore enthusiast may buy a cheap cabinet to mod, and a powerful power supply.

More fans means the cabinet is better: While more fans may seem to aid ventilation, more critical is the airflow. It’s not the number of fans that is important. It’s the direction of airflow, as is the ability of a cabinet’s fans to draw in fresh (cooler) air, and expel hot air simultaneously.

QUESTIONS TO ASK

Does the cabinet support at least two fans?

You’ll need two fans at least, one to remove hot air, and another to blow in cool air. Your cabinet may not come with two fans, but should have the required screw holes allowing you to mount such fans later on. As mentioned, try to ensure support for larger — 120 mm fans.

I’ve heard of “compliance” in relation to cabinets. What’s that?

The latest cabinets sport compliances to certain thermal standards. Because these standards are the “minimum

recommended,” some enthusiast cabinets may exceed these standards. In such cases compliances will not be mentioned. However, the TAC standard (Thermally Advantaged Chassis) is one such standard that is pretty much adhered to (or exceeded) by most cabinet manufacturers. The TAC standard calls for a compulsory vent in the side panel over the processor-heatsink area, called the CAG (Chassis Air Guide). TAC also makes mandatory another, smaller vent above the graphics card area. Besides this, front and rear vents (which are capable of being populated by fans)



are necessitated. The reason behind the TAC standard was the excessive heat generation by today’s PCs, which causes stability issues especially in tropical climates such as ours. It’s good to look out for a cabinet that has a few (if not all) of these features. Give larger fans preference: they’re quieter and move more air.

Future Trends

Cabinets don’t “evolve.” Rather, they react to changes in other product categories. We’ve seen a lot of snazzy “see-thru” cabinets, meaning the presence of a transparent, windowed side panel. Manufacturers aren’t very far behind the case modding enthusiasts with the jazz factor, and the good news is when you go cabinet shopping, there’s absolutely no dearth of good choices. ■



What To Look For

Form Factor: Cabinets are compatible with all ATX motherboards, so your choice of cabinet will never restrict your choice of hardware.

Desk space is your main deterrent here. The basic form factors are Full Tower or just Tower (tall, full-size cabinets), Mid-Tower (which typically support 4 x 5.25 inch external drive bays), and Mini-Tower.

As a rule, cabinet vendors aren’t very vocal about these standards. You’ll need to do your homework here.

Expandability: A cabinet will typically house three to five optical drives, depending on its height (form factor). Always look at the word “external” when checking on the number of available optical bays. A cabinet may support six 5.25-inch drives but only four may be external. This virtually means that the two internal 5.25-inch drive bays are useless, unless the cabinet bundle includes a 5.25-inch to 3.5-inch converter (in which case these are usable as hard drive bays). Similarly, for hard drives, you need to check for the number of internal 3.5-inch bays.